

EVERY GRAPH WITH NO K_7^- MINOR IS SIX-COLOURABLE

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ABSTRACT. Let K_7^- be the graph obtained from K_7 by deleting one edge. We prove that every graph with no K_7^- minor is six-colourable. The main extremal result is that every 4-bilight graph on $n \geq 3$ vertices with at least $4n - 2$ edges contains a K_7^- minor. The proof strengthens the rooted density construction of Dvořák, Norin and Rahman: five prescribed root bags admit two additional bags with at most one missing contact. A separator argument then reduces the extremal theorem to six-connected graphs. The final degree-seven case uses a finite verification of 232 nine-vertex graphs.

1. INTRODUCTION

All graphs in this paper are finite and simple. Write K_7^- for K_7 with one edge deleted. We prove the following statement, posed as Conjecture 21 by Norin and Totschnig [5].

Theorem 1.1. *Every graph with no K_7^- minor is six-colourable.*

Norin and Totschnig [5] proved the corresponding result when the excluded minor is K_7 with two incident edges deleted. Dvořák, Norin and Rahman [2] proved it for K_7 with two independent edges deleted. Theorem 1.1 strengthens both conclusions. It does not settle Hadwiger's conjecture for K_7 .

The proof builds on the rooted density and separation framework of Dvořák, Norin and Rahman. Their critical-graph reduction turns Theorem 1.1 into an extremal problem. Our main result is Theorem 2.1, for the class of 4-bilight graphs defined below. In particular, it proves their Conjecture 1.5:

Corollary 1.2. *Every five-connected graph on $n \geq 6$ vertices with at least $4n - 2$ edges contains a K_7^- minor.*

The new rooted construction gives five prescribed root bags and two root-free bags, with at least ten of the eleven possible contacts involving the latter two bags. It uses a rooted-star construction, careful contraction across five-separations, and Mader's atom theorem. The global argument first excludes vertices of degree five and then all five-cuts. In the resulting six-connected graph, contractions force dense degree-six and degree-seven neighbourhoods, which yield the minor.

We state the external tools precisely in Section 2 and supply the supporting constructions and the full global argument in the subsequent sections. The only computational input is Lemma A.1; Appendix A explains its finite scope, certificates and reproducibility.

2. NOTATION AND EXTERNAL TOOLS

2.1. Minors and densities. Write $v(G) = |V(G)|$, $e(G) = |E(G)|$ and $\rho(G) = e(G) - 4v(G)$. A minor model consists of pairwise disjoint, nonempty connected vertex sets, called *bags*, with an edge between bags whenever the target requires one. In a graph with prescribed roots X , a rooted model retains each root in its corresponding bag, and no bag contains two roots. Extra edges are allowed. We allow an empty root set and interpret its rooted clique K_0 as the empty model.

For a nonempty set $Y \subseteq V(G)$, let $N_G(Y)$ be its external neighbourhood, and put

$$\rho_G(Y) = e(G[Y]) + e_G(Y, N_G(Y)) - 4|Y|.$$

We call Y a *fragment*, its neighbourhood its *boundary*, and $V(G) \setminus (Y \cup N_G(Y))$ its *opposite side*. A fragment is *proper* if its opposite side is nonempty. In a rooted graph, fragments are required to avoid the roots. Set

$$\rho_4(G, X) = e(G) - e(G[X]) - 4|V(G) \setminus X|.$$

Thus $\rho_4(G[Y \cup N_G(Y)], N_G(Y)) = \rho_G(Y)$.

A k -rooted graph (G, X) is *4-light* if $\rho_G(Y) \leq 0$ for every root-free fragment with fewer than k boundary vertices. In particular, for five roots the tested boundary orders are at most four. The graph is *internally k -connected* if every nonempty root-free set has at least k external neighbours.

Two disjoint, nonadjacent fragments form a *bifragment*; it is *dense* if both fragments have positive density. Here nonadjacent means that no edge joins the two fragments. An unrooted graph is *4-bilight* if it has no dense bifragment with both boundary orders at most four. Every five-connected graph is 4-bilight. Our main extremal theorem is as follows.

Theorem 2.1. *Every 4-bilight graph G with $v(G) \geq 3$ and $e(G) \geq 4v(G) - 2$ contains a K_7^- minor.*

A separation (A, B) has $A \cup B = V(G)$ and no edge between $A \setminus B$ and $B \setminus A$; its order is $|A \cap B|$. It is proper if both open sides are nonempty. For a root separation we also require $X \subseteq A$. Its right-hand side is $G[B]$ rooted at $A \cap B$. When a linkage carries these boundary roots back to X , its paths are disjoint and their interiors avoid B . Roots already on the boundary use trivial paths. Appending the paths to their respective bags preserves disjointness and all contacts.

We repeatedly use the following lifting convention. Every vertex of a minor is assigned its fixed connected preimage in the original graph. Preimages of different vertices are disjoint. Replacing a bag by the union of its preimages lifts connectivity, contacts and all prescribed roots. Our minimal-counterexample arguments minimise $(v(G), e(G))$ lexicographically; each reduction explicitly preserves the relevant density and separation conditions.

2.2. Rooted density theorems. Let $\mathcal{S}_{5,j}$ be the class of graphs on five vertices with at most j edges, and let $\mathcal{S}_{5,4}^-$ omit $K_3 \dot{\cup} K_2$ from $\mathcal{S}_{5,4}$. The following form of Dvořák's theorem [1, Theorem 4] is reproduced in [2, Theorem 2.6].

Theorem 2.2. *If (G, X) is a 4-light five-rooted graph and $t = \rho_4(G, X)$, then it has a rooted model of every labelled graph in $\mathcal{T}_t$, where*

t	≤ 0	1	2	3	4	5	6	≥ 7
$\mathcal{T}_t$	$\mathcal{S}_{5,0}$	$\mathcal{S}_{5,1}$	$\mathcal{S}_{5,3}$	$\mathcal{S}_{5,4}^-$	$\mathcal{S}_{5,6}$	$\mathcal{S}_{5,8}$	$\mathcal{S}_{5,9}$	$\mathcal{S}_{5,10}$

The labelling prescribes the root for each target vertex.

Corollary 2.3 ([2, Lemma 5.6]). *Let (G, X) be 4-light with five roots, and put $m = 10 - e(G[X])$. If $\rho_4(G, X) \geq \lceil (m + 3)/2 \rceil$, then G has a rooted K_5 minor.*

Proof. The target class at this threshold, and every higher threshold, contains every graph with m edges. Apply Theorem 2.2 to the missing edges of $G[X]$, then retain the existing root edges. $\square$

A *dart* has four roots, two root edges forming a three-vertex path, and one nonroot adjacent to all four roots. The fourth root is isolated in the root-induced graph. Dvořák's small-root theorem [1, Corollary 13], together with [2, Lemma 3.1], gives the following fact: a 4-light k -rooted graph of positive rooted density contains a rooted K_k if $k \leq 3$, and a rooted dart if $k = 4$.

2.3. Reducible fragments and extraction. A fragment Y with boundary S of order at most four is *reducible* if $\rho_G(Y) \leq 0$ and $G[Y \cup S]$, rooted at S , contains either a rooted clique on S , or a rooted dart when $|S| = 4$ and $|Y| \geq 2$. For an unrooted ambient graph we also require $|V(G) \setminus Y| \geq 3$.

A *reducent* replaces Y by such a clique or dart, retaining the graph outside Y . The replacement is a minor: the rooted model provides fixed preimages for the boundary vertices and, for a dart, its one additional vertex. In both cases the order strictly decreases.

Lemma 2.4 ([2, Lemma 3.3]). *Replacing an inclusionwise-maximal reducible fragment preserves 4-bilightness and does not decrease ρ in an unrooted 4-bilight graph. In a 4-light five-rooted graph, it preserves 4-lightness and does not decrease ρ_4 . All original roots are retained in distinct prescribed bags.*

For the extraction step, two weaker patterns from [2] are useful. Each has five independent roots and two nonroots. In the first, the nonroots are adjacent and each misses at most one root, with distinct missed roots if both omissions occur. In the second, the nonroots are nonadjacent and both meet every root; this is $K_{2,5}$ rooted at its five-vertex part.

Lemma 2.5 ([2, Corollary 4.2]). *Let (G, X) be 4-light, with $|X| \leq 4$ and $\rho_4(G, X) \leq 0$. Suppose a root five-separation (A, B) has a right-hand side containing one of the two weaker patterns. Then G has a reducible fragment Y containing $B \setminus A$.*

The mechanism, also used for five roots below, is worth specifying. Restrict either pattern to a proper subset Z of its roots by deleting the other root vertices. The remaining graph contains a rooted clique if $|Z| \leq 3$, or a dart if $|Z| = 4$ [2, Observation 4.1]. If five disjoint paths do not link the original roots to the boundary, Menger's theorem gives an isolating separator of order at most four, linked to a proper subset of the boundary. The clique or dart lifts along this linkage. Its interior has at least two vertices, because the two helper bags remain there. Lightness gives nonpositive density when the isolator is smaller than the root set. When its order equals the root count, choose the isolator $(X, V(G))$; its density is $\rho_4(G, X) \leq 0$ by assumption. This gives the reducible fragment. When transferring it to an unrooted ambient graph, we separately check that at least three vertices remain outside it.

We also use the root-forgetting rule [2, Observation 5.2]: if $|X| \leq 4$, (G, X) is 4-light and $\rho_4(G, X) \leq 0$, then the same is true after replacing X by any subset X' . To see the density bound, put $M = X \setminus X'$. For a set Y tested in the new rooted graph, or its entire nonroot set, put $Y' = Y \setminus M$. The old hypotheses give $\rho_G(Y') \leq 0$, interpreting the empty set as having density zero. At most $4|M \cap Y|$ additional incident edges are counted when $M \cap Y$ is restored: each such vertex has possible neighbours, outside Y' , only in $M \cap Y$ or $N_G(Y)$, and $|M| + |N_G(Y)| \leq 4$. Hence

$$\rho_G(Y) \leq \rho_G(Y') + 4|M \cap Y| - 4|M \cap Y| \leq 0.$$

3. LOCAL ROOTED CONSTRUCTIONS

We first record three small rooted constructions and a local contraction lemma. They will also be used in the unrooted argument.

Lemma 3.1 (Rooted star). *Let $Z \subseteq V(H)$ have order k , and let $X \subsetneq Z$ be independent. Suppose that*

$$|V(H)| + |X| \leq 2k - 1, \quad d_H(v) \geq k - 1 \quad (v \notin X), \quad d_H(x) \geq k - |X| \quad (x \in X).$$

Then, for some $z \in Z \setminus X$, the set $(V(H) \setminus Z) \cup \{z\}$ is connected and adjacent to every vertex of $Z \setminus \{z\}$.

Proof. Put $T = V(H) \setminus Z$. A component D of $H[T]$ misses at most $|D|$ vertices of Z : any vertex of D has at least $k - |D|$ neighbours in Z . Consequently at most $|T|$ roots miss some component of $H[T]$. Since $|T| + |X| \leq k - 1$, there is a vertex $z \in Z \setminus X$ adjacent to every component. Thus $T \cup \{z\}$ is connected, including when $T = \emptyset$.

A vertex $r \in Z \setminus \{z\}$ which has no neighbour in T must be adjacent to z . For $r \notin X$ this follows from $d_H(r) \geq k - 1$; for $r \in X$, independence of X and $d_H(r) \geq k - |X|$ force adjacency to every vertex of $Z \setminus X$. $\square$

Lemma 3.2 (Rooted triangle). *Let (R, S) be internally three-connected, where $|S| = 3$. Suppose that $R - S$ is nonempty and connected, each root has a neighbour outside S , and every nonroot has degree at least four. Then R contains an S -rooted triangle minor.*

Proof. Three distinct vertices a, b, c of a two-connected graph have a rooted triangle minor. To see this, take a cycle through a, b and, if necessary, a two-fan from c to that cycle. Either their union contains a cycle through all three vertices, or it contains a theta whose three paths contain a, b, c separately. In the latter case, assign one end of the theta to the a -path and the other to the b -path; the interior of the third path is the c -bag. These three bags are connected and pairwise adjacent.

Consider the block-cut tree of R , representing a root which is a cutvertex by its cutvertex node, and any other root by its block node. If the median of the three root nodes is a block, the roots have three distinct attachments to that block. The block is two-connected, so the preceding observation supplies a triangle rooted at the attachments. Appending the three disjoint exterior root paths gives the result.

If the median is a cutvertex c , each component of $R - c$ contains at most one root. The nonroots in any such component would have boundary contained in c and that possible root, contrary to internal three-connectivity. Hence c is the only nonroot, but then its degree is at most three. This is also impossible. $\square$

We use the following terminal construction from the proof of [2, Lemma 3.1]. Assume that four independent roots are the leaves of a tree T . Its two degree-three branching vertices v_1, v_2 may coincide at a degree-four vertex. Choose T to minimise the length of the path between these vertices. If, after deleting v_1, v_2 , every partition of the roots into two pairs has a path joining the pairs, then the graph has a rooted dart. This part of the cited proof uses only the stated tree and path conditions, not a density hypothesis.

Lemma 3.3 (Rooted dart). *An internally four-connected graph with four prescribed roots, at least two nonroots, and nonroot minimum degree at least five contains a rooted dart.*

Proof. Write (F, Z) for the rooted graph and induct on $|V(F)|$. Delete the root-root edges; this preserves all nonroot degrees and boundaries. Every partition of Z into two pairs has two disjoint paths between the pairs. Otherwise Menger's theorem gives a separation of order at most one, with one pair on each side. The nonroots on either open side then have boundary contained in that pair and the separator, of order at most three. Internal four-connectivity would put all nonroots in the separator, contradicting their number.

Each component of $F - Z$ sees all four roots. It therefore supplies a tree whose leaves are exactly Z . Choose such a tree with a shortest branching path, with ends v_1, v_2 . Unless $\{v_1, v_2\}$ separates two roots from the other two, the terminal construction above gives a dart. The case $v_1 = v_2$ is covered, since a one-vertex separation of this kind was already excluded.

In the remaining case let (B_1, B_2) be the separation, with $B_1 \cap B_2 = \{v_1, v_2\}$ and two roots in each open side. Put

$$Z_i = (Z \cap B_i) \cup \{v_1, v_2\}, \quad W_i = B_i \setminus Z_i.$$

If both W_i are empty, the degree bound makes v_1, v_2 adjacent to each other and to every root. Use v_1 as the dart's helper and absorb v_2 into one root bag: the resulting root star contains the required three-vertex path.

Otherwise choose a nonempty W_i . It has at least two vertices, since a single vertex would have at most the four neighbours in Z_i . Every subset of W_i has the same neighbours in $F[B_i]$ as in F . Thus $(F[B_i], Z_i)$ is internally four-connected and retains the nonroot degree bound. It omits the two opposite roots, so induction provides a Z_i -rooted dart.

Take the two disjoint paths between the two original root pairs established above. They cross the separator at different vertices. Their segments after their last visits to $\{v_1, v_2\}$ lie outside

B_i , apart from their initial vertices, and end at the two opposite roots. Append these segments to the bags at v_1, v_2 . They are disjoint from the other bags and from the helper, and restore the four original roots without losing any contact. $\square$

Call a graph *quasi five-connected* if it is four-connected and has no four-separation with at least two vertices on each open side. The following local version of the crossing argument in [3, Lemma 1] does not assume contraction-criticality.

Lemma 3.4. *Let Q be five-connected, and let A have boundary $S = N_Q(A)$ of order five. Suppose that $|A| \geq 3$ and $|V(Q) \setminus (A \cup S)| \geq 2$. If $x \in S$ has a unique neighbour a in A , then Q/xa is quasi five-connected.*

Proof. An edge contraction preserves four-connectivity. A forbidden four-separation of Q/xa would lift to a five-cut T of Q containing x, a , with open sides B, D of order at least two. Write $C = V(Q) \setminus (A \cup S)$ and

$$p = |A \cap T|, \quad r = |C \cap T|, \quad s = |S \cap T|, \quad u = |S \cap B|, \quad w = |S \cap D|.$$

Then $p + r + s = u + w + s = 5$ and $s \geq 1$.

If $A \cap B \neq \emptyset$, its boundary is contained in $(A \cap T) \cup (S \cap T) \cup (S \cap B)$ and omits x , whose unique neighbour in A lies in T . Five-connectivity gives $p + s + u \geq 6$, so $p > w$ and $u > r$. The opposite corner $C \cap D$ has boundary of order at most $r + s + w < 5$, and is therefore empty. Symmetrically, $A \cap D \neq \emptyset$ implies $p > u$, $w > r$, and $C \cap B = \emptyset$.

If both A -corners are nonempty, then $C \subseteq T$, so $r \geq 2$, whereas $u + w \geq 2r + 2 \geq 6 > 5 - s$. If neither is nonempty, then $p = |A| \geq 3$ and $r \leq 1$. Some C -corner is nonempty, say $C \cap B$; its boundary gives $u \geq p$. Hence $w \leq 1$ and $C \cap D$ is empty, contrary to $|D| \geq 2$.

Finally suppose only $A \cap B$ is nonempty. Then $D = S \cap D$, so $w \geq 2$. Since $p > w$, we have $r \leq 1$. But $C \cap B$ has boundary of order at most $r + s + u = 5 + r - w \leq 4$, so it too is empty. This leaves $|C| = r \leq 1$, the final contradiction. $\square$

4. TWO ROOTED HELPERS

For five prescribed roots X , say that a model has the *helper property* if it consists of five prescribed-root bags and two further nonempty connected bags, all pairwise disjoint, and at least ten of the eleven possible helper-root and helper-helper contacts are present. The two helpers contain no root; no root-root contact is required.

Theorem 4.1. *Every 4-light five-rooted graph (G, X) with $\rho_4(G, X) \geq 2$ has the helper property.*

The proof first restricts a minimum counterexample. It then uses a weighted neighbourhood to find an edge with few common neighbours. An auxiliary clique allows Mader's atom theorem to be applied without requiring intersections of positive-density fragments to remain positive.

4.1. Minimum-counterexample reductions. Suppose the theorem is false, and choose a counterexample minimising $(|V(G)|, |E(G)|)$ lexicographically. Throughout this section, contractions identify at most one original root in each preimage. A rooted model in a minor lifts by replacing each of its vertices with the fixed connected preimage; in particular, all five prescribed roots remain distinct.

We use the following consequence of the isolator argument in [2, Observation 4.1 and proof of Corollary 4.2]. Suppose a root-free region has a helper-property model on a five-vertex boundary, or two helpers full to a larger boundary. Either five disjoint exterior paths attach the original roots to five boundary bags, or a separator of order $k \leq 4$ isolates the region and has a linkage to k of its boundary bags. A helper model restricted to those k bags contains a rooted K_k for $k \leq 3$, and a rooted dart for $k = 4$. Indeed, retain a helper adjacent to all four roots and absorb the other into a root bag to obtain two root contacts. For three roots, absorb the helpers into two root bags; smaller cases follow by restriction. The same conclusion holds when both helpers are full to a larger boundary. If the isolated interior has at least two vertices and nonpositive density, it is a reducible fragment. The linkage is outside the original region, except for its ends, so it does not use a helper vertex. This is the only form of the isolator argument needed below.

Lemma 4.2. *The minimum counterexample has the following properties:*

- (1) *X is independent, and there is no reducible root-free fragment with boundary of order at most four.*
- (2) *Every proper root-free set Y with $|N_G(Y)| = 5$ satisfies $\rho_G(Y) \leq 1$.*
- (3) *$G - X$ is connected, every nonroot has at most three root neighbours, and every root has degree at least two.*
- (4) *$\rho_4(G, X) = 2$.*

Proof. Root-root edges contribute neither to the density nor to a required helper contact, and cannot occur inside a bag containing just one root. Thus minimality makes X independent. A maximal reducible fragment would give, by [2, Lemma 3.3], a smaller 4-light rooted minor of nondecreasing density. Its helper model would lift, proving (1).

For (2), put $S = N_G(Y)$. The induced rooted graph $(G[Y \cup S], S)$ is 4-light, since its root-free sets retain their incident edges and neighbours. If its density is at least two, it is a smaller graph with a helper model by minimality. Five exterior root paths would extend that model to X . Otherwise the isolator argument gives a reducible fragment: its interior contains the two helpers and has nonpositive density by 4-lightness. Both possibilities contradict (1).

The densities of the components of $G - X$ sum to $\rho_4(G, X)$. Each positive-density component sees all five roots, by 4-lightness. Two such components give two full helpers, with only their mutual contact possibly missing. Hence there is exactly one positive component and its density is at least two. Any other component would make this a proper five-boundary set, contrary to (2). Thus $G - X$ is connected and sees every root.

Suppose a nonroot z has $d \geq 4$ root neighbours. For the components Q_i of $G - (X \cup \{z\})$,

$$\sum_i \rho_G(Q_i) = \rho_4(G, X) + 4 - d.$$

Each Q_i sees z , and each positive one sees at least four roots. If $d = 5$, there is a positive component; it and $\{z\}$ are adjacent helpers with at least ten contacts. If $d = 4$, let x be the missing root. A component seeing all roots already gives the required model. Otherwise each positive component has exactly four root neighbours, and has density at most one by (2). Their sum is at least two, so there are at least two positive components. Choose a component R seeing x , and a positive component $Q \neq R$. The helpers $\{z\} \cup R$ and Q are adjacent and have respectively five and at least four root contacts. This again contradicts the choice of G .

If a root x had just one neighbour v , the side $G - x$, rooted at $(X \setminus \{x\}) \cup \{v\}$, would have density $\rho_4(G, X) + 4 - |N_G(v) \cap X| \geq 3$. By 4-lightness, all five proposed boundary vertices must be actual neighbours of its interior. It is therefore a proper five-boundary side, contradicting (2). This proves (3).

Finally suppose $\rho_4(G, X) \geq 3$ and delete an edge with a nonroot end. The result has density at least two but, by minimality, is not 4-light. A positive small-boundary set Y becomes permissible only because the deleted edge joins Y to a vertex outside its closed side. Restoring that edge gives a five-boundary set of density at least two. By (2) its closed side is all of G , and its boundary is X . The restored root endpoint then has only one neighbour, contrary to (3). Therefore the density is exactly two. $\square$

Lemma 4.3. *Every nonroot has degree at least five, and (G, X) is internally five-connected.*

Proof. Deleting a nonroot v of degree at most four leaves density $2 + 4 - d_G(v) \geq 2$. If $G - v$ has a positive root-free set Y with at most four neighbours, restoring v must add a neighbour to Y ; otherwise Y already violates 4-lightness in G . Its restored density is at least two and its boundary has order five. This is a proper side, since its boundary contains the nonroot v and could not also contain all five roots. Lemma 4.2(2) excludes it. Thus deletion stays in the induction class, a contradiction.

If internal five-connectivity fails, choose a connected root-free set Y of minimum boundary order $k \leq 4$, and put $S = N_G(Y)$. The rooted shore $(G[Y \cup S], S)$ is internally k -connected, retains all nonroot degrees, and has nonpositive density. For $k = 0, 1, 2$, it contains a rooted K_k (using a path through Y when $k = 2$). For $k = 3$, apply Lemma 3.2. For $k = 4$, the

degree bound implies $|Y| \geq 2$, and Lemma 3.3 applies. In every case Y is reducible, contrary to Lemma 4.2(1). $\square$

Complete X to a clique, obtaining $Q_0 = G + K_X$. It is five-connected: after deleting at most four vertices the surviving roots lie in one clique, and another component would contradict internal five-connectivity. For an original edge e having a nonroot end, let $t(e)$ be its number of common neighbours in Q_0 . The rooted density after contraction is

$$(1) \quad \rho_4(G/e, X_e) = 2 + 3 - t(e),$$

where X_e denotes the unchanged or naturally identified roots. The completion counts exactly the edges which become root–root edges when one endpoint is a root.

Lemma 4.4. *If $e = uv$ has a nonroot end and $t(e) \leq 3$, there is a proper root-free set Y such that*

$$|N_G(Y)| = 5, \quad \rho_G(Y) = 1, \quad \{u, v\} \subseteq N_G(Y),$$

and no vertex of Y is adjacent to both u and v .

Proof. By (1) and minimality, G/e is not 4-light. Let Y be a positive root-free set witnessing this. The identified vertex cannot lie outside its closed side, because then Y is a root-free set of boundary at most four in G . Nor can it lie inside Y : its preimage would have the same small boundary (and would be root-free; a contracted root cannot be interior). It therefore lies on the boundary. Splitting it gives exactly five neighbours in G . The side is proper, since its boundary contains a nonroot and hence cannot contain all five original roots. If c vertices of Y see both endpoints, then $\rho_G(Y) = \rho_{G/e}(Y) + c$. Lemma 4.2(2) forces both densities to equal one and $c = 0$. $\square$

Lemma 4.5. *There are no adjacent degree-five nonroots with exactly three common neighbours.*

Proof. Suppose p, q are such vertices. Write their common neighbours as the three-set A , and their respective exclusive neighbours as u, v . Lemma 4.4 applied to pq gives a density-one set T with boundary $\{p, q\} \cup D$, where $|D| = 3$, avoiding A . Since both p, q have neighbours in T , their neighbourhoods force $u, v \in T$; in particular these are nonroots.

Delete p, q from this shore and reroot at $\{u, v\} \cup D$:

$$R = G[T \cup D], \quad Z = \{u, v\} \cup D.$$

The graph (R, Z) is 4-light. Indeed, every subset of its nonroots $T \setminus \{u, v\}$ has exactly the same neighbours and incident edges as in G , since only u, v in T see p, q . Put $a_0 = e_G(\{u, v\}, D)$ and $\delta = 1$ if $uv \in E(G)$, and $\delta = 0$ otherwise. Then

$$\rho_4(R, Z) = (4|T| + 1 - 2 - \delta - a_0) - 4(|T| - 2) = 7 - \delta - a_0.$$

This is exactly the number $m \in \{0, \dots, 7\}$ of missing edges from uv together with the six edges between $\{u, v\}$ and D . The rooted universality theorem [2, Theorem 2.6] supplies all these missing edges simultaneously: its table contains every m -edge graph in this range (the exceptional graph at density three has four edges). Retain the root edges already present. We obtain five rooted bags in R , with the bags at u, v adjacent to one another and to all three bags at D .

Append p to the u -bag and q to the v -bag. These are adjacent root-free helpers, full to the boundary $A \cup D$. For $A \cap D$ use the existing D -bags; for $A \setminus D$ use singletons. These choices are disjoint because $T \cap A = \emptyset$.

The root-free set $U = T \cup \{p, q\}$ has boundary exactly $A \cup D$. If five exterior paths link X to this boundary, retain the five bags at their distinct ends and append the paths. This gives the helper property rooted at X . Otherwise the isolator argument gives a separator of order at most four, a rooted clique or dart on it, and a root-free interior containing U . That interior has at least two vertices and nonpositive density, so is reducible. Both alternatives are impossible. $\square$

4.2. Eliminating degree five.

Lemma 4.6. *If u, v are nonroots and Q_0/uv is quasi five-connected, then $(G/uv, X)$ is 4-light.*

Proof. Root completion does not change any root-free boundary or incident density. Every root has degree at least five in Q_0/uv : its at least two nonroot neighbours merge to at least one, and it retains the four other roots as neighbours. A root-free set with boundary at most four has nonempty opposite side, because there are five roots. Its boundary therefore has order four, and quasi five-connectivity makes one open side a singleton. A singleton root-free side has degree four and density zero. A singleton opposite side would be a root of degree at most four, which is impossible. $\square$

Lemma 4.7. *Every nonroot of G has degree at least six.*

Proof. Suppose a nonroot v has degree five. First assume $Q_0[N_G(v)]$ is not complete, and choose a nonedge ab in it. Its ends are not both roots. The graph

$$H = G - v + ab$$

has rooted density two and is a proper rooted minor: contract va and delete the unwanted new edges. If H is 4-light, its helper model lifts, contrary to minimality.

Otherwise take a positive root-free set Y of H with at most four neighbours. Put $k = e_G(v, Y)$ and let c indicate whether the added edge ab is incident with Y . Then

$$N_G(Y) \subseteq N_H(Y) \cup \{v\}, \quad \rho_G(Y) = \rho_H(Y) + k - c.$$

Internal five-connectivity gives $|N_G(Y)| = 5$ and $k \geq 1$. This side is proper, since its boundary contains the nonroot v . Consequently

$$1 \geq \rho_G(Y) = \rho_H(Y) + k - c \geq 1 + k - c,$$

forcing $\rho_G(Y) = \rho_H(Y) = k = c = 1$. Exactly one endpoint of ab lies in Y ; call it a . Then a is a nonroot and the unique neighbour of v in Y . The other endpoint b is among the four vertices of $N_G(Y) \setminus \{v\}$, since the added edge cannot introduce a fifth neighbour in H .

The set Y is not a singleton, which would make ab an edge of G . If $|Y| = 2$, its density one and the nonroot degree bound force adjacent degree-five vertices with three common neighbours, contrary to Lemma 4.5. Hence $|Y| \geq 3$. Its opposite side in Q_0 has at least two vertices: a singleton would be an original root z , with boundary $(X \setminus \{z\}) \cup \{v\}$, and independence of X would give $d_G(z) \leq 1$.

Lemma 3.4 now applies to the unique edge va into Y . Thus Q_0/va is quasi five-connected, and Lemma 4.6 makes G/va 4-light. Since v has degree five and b misses a , the edge va has at most three common neighbours. Its contraction retains rooted density at least two and decreases order, a contradiction.

It remains that $Q_0[N_G(v)]$ is a clique. Internal five-connectivity and Menger's theorem give five disjoint paths from X to $N_G(v)$, with interiors outside $N_G[v]$. Roots already in $N_G(v)$ use trivial paths. Choose $z \in N_G(v) \setminus X$, which exists because v has at most three root neighbours. Use $\{v\}, \{z\}$ as helpers, and the five linkage paths as root bags, removing z from its own path. That trimmed bag is nonempty because z is not an original root. The helper z meets all five root bags, v meets the other four, and vz joins the helpers. Every used clique edge has nonroot end z , so is an actual edge of G . This gives the ten required contacts. $\square$

4.3. Weighted neighbourhoods and an atom. Let E_0 be the original edges with a nonroot end and at most three common neighbours in Q_0 . Each edge of E_0 has a five-boundary set as in Lemma 4.4.

Lemma 4.8. *If v is a nonroot and $d_G(v) + |N_G(v) \cap X| \leq 9$, then some edge of E_0 is incident with v .*

Proof. Suppose instead that $t(vu) \geq 4$ for every neighbour u of v . Put $H = G[N_G(v)]$, $X_0 = N_G(v) \cap X$, and $r = |X_0|$. Vertices of $H - X_0$ have degree at least four in H ; vertices of X_0

have degree at least $5 - r$, since root completion adds exactly $r - 1$ common neighbours to such an edge. Also X_0 is independent and $|V(H)| + r \leq 9$.

There are five disjoint paths from X to $N_G(v)$ in $G - v$. Otherwise a separator of order at most four would isolate, in G , a root-free component containing v . Truncate the paths at their first entries to $N_G(v)$. Their five ends form a set Z containing X_0 , and their interiors avoid the neighbourhood. Lemma 3.1 gives a central bag

$$C = (N_G(v) \setminus Z) \cup \{z\}, \quad z \in Z \setminus X_0,$$

adjacent to the four other ends. Use C and $\{v\}$ as helpers, keeping the four other linkage paths and trimming z from its path. The trimmed root bag is nonempty; all bags are disjoint. The helper C has five root contacts, v has four, and they are adjacent. This is the helper property, a contradiction. $\square$

In particular, E_0 is nonempty. If $m = |V(G) \setminus X|$, the nonroot degree bound and at most three root neighbours give $m \geq 4$, while

$$(2) \quad \sum_{v \notin X} (d_G(v) + |N_G(v) \cap X|) = 8m + 4.$$

Some nonroot therefore has weight at most nine.

We use Mader's atom theorem in the form stated in [4, Theorem 5]. In a graph Q of connectivity k , let $\mathcal{S}$ be any family of vertex sets. An $\mathcal{S}$ -fragment is a nonempty proper union of components of $Q - T$, where T is a k -cut containing a member of $\mathcal{S}$. An $\mathcal{S}$ -atom is such a fragment of minimum order. For an atom A , put $T_A = N_Q(A)$ and $\bar{A} = V(Q) \setminus (A \cup T_A)$. If some $S \in \mathcal{S}$ and k -cut T satisfy

$$(3) \quad S \subseteq T \setminus \bar{A}, \quad A \cap T \neq \emptyset,$$

then

$$(4) \quad A \subseteq T, \quad |A| \leq \frac{1}{2}|T \setminus T_A|.$$

There is no assumption that every edge is noncontractible.

Completion of the proof of Theorem 4.1. Add a set W of $|V(G)| + 1$ vertices and make $X \cup W$ a clique, with no edges from W to $V(G) \setminus X$; call the resulting graph Q . Internal five-connectivity shows that Q is five-connected, and X separates $G - X$ from W , so its connectivity is exactly five. This padding changes neither boundaries of original root-free sets nor common-neighbour counts of original edges with a nonroot end.

Let $\mathcal{S} = \{V(e) : e \in E_0\}$. By Lemma 4.4, $\mathcal{S}$ -fragments exist in Q , with an example of order at most $m = |V(G) \setminus X|$. Choose an $\mathcal{S}$ -atom A , among all such fragments, so $|A| \leq m$. It avoids $X \cup W$: otherwise its five-boundary would force it to contain the whole surviving clique, of order at least $|W| > m$. Thus $A \subseteq V(G) \setminus X$, and its boundary is the same in G and Q .

Its closed side is proper in G . If not, its five-boundary would be exactly X , and could not contain the ends of an edge in E_0 . Lemma 4.2(2) therefore gives $\rho_G(A) \leq 1$. The nonroot degree bound excludes $|A| = 1, 2$, since for these orders

$$\rho_G(A) \geq 2|A| - \binom{|A|}{2} \geq 2.$$

Writing $S_A = N_G(A)$, double-counting incident edges gives

$$\sum_{v \in A} (d_G(v) + |N_G(v) \cap S_A|) = 8|A| + 2\rho_G(A) \leq 8|A| + 2.$$

As $|A| \geq 3$, some vertex $v \in A$ has the displayed weight at most eight. Its root weight is no larger, so Lemma 4.8 supplies an incident edge $f \in E_0$. Both ends of f lie in $A \cup T_A$.

Apply Lemma 4.4 to f , and let T be the boundary of its resulting fragment. In Q , this is a five-cut containing both ends of f and meeting A at v . Consequently $V(f) \subseteq T \setminus \bar{A}$, so (3) holds. Mader's theorem yields $|A| \leq 5/2$, contradicting $|A| \geq 3$.

The auxiliary clique has only selected an atom: no model has been constructed in the padded graph. The contradiction proves the theorem in the original rooted graph. $\square$

5. ORIENTING THE SEPARATIONS

Suppose that Theorem 2.1 is false, and choose a counterexample G minimising $(v(G), e(G))$ lexicographically. Thus G is 4-bilight, has at least three vertices and $\rho(G) \geq -2$, and has no K_7^- minor. Simplicity gives $v(G) \geq 9$. Throughout the next two sections, G denotes this minimum counterexample. Every use of minimality will preserve 4-bilightness, the density bound and the lower bound of three on the order. Models in a minor lift by replacing each contracted vertex with its fixed connected preimage.

For a fragment A , write R_A for $G[A \cup N_G(A)]$ rooted at $N_G(A)$. We call a five-rooted side *heavy* if its rooted density is at least two, and *low* if its density is at most one. For a separation (A, B) , write $R_{A,B} = G[B]$, rooted at $A \cap B$.

Lemma 5.1. *The graph G has no reducible fragment.*

Proof. Replace an inclusion-maximal reducible fragment by its boundary-rooted clique or dart. By Lemma 2.4, the resulting graph is a minor, is 4-bilight and has density at least $\rho(G)$. The definition of reducibility leaves at least three vertices outside the fragment. The replacement strictly decreases the order: a clique removes every interior vertex, and a dart retains one when there were at least two. This contradicts minimality. $\square$

The next lemma adapts the separation argument of [2, Section 5]. We give the details because the threshold two and the final K_7^- model are important here.

Lemma 5.2. *The following assertions hold.*

- (1) *If one side of a five-separation is heavy, the opposite side is 4-light.*
- (2) *For every five-separation with boundary X , exactly one side is heavy. If $m = 10 - e(G[X])$, the heavy side has density at least $m + 7$.*
- (3) *There is no bfragment (S, T) with both boundaries of order at most five, both densities positive, and every member with boundary of order five heavy.*

Proof. We first record the small-separation step. Suppose (C, D) has order at most four, $R_{C,D}$ is 4-light and has nonpositive density, and

$$|(C \setminus D) \cup N_G(C \setminus D)| \geq 3.$$

There cannot be a five-separation (A, B) with $B \subseteq D$ and $R_{A,B}$ both 4-light and heavy. To see this, remove from $C \cap D$ the vertices with no neighbour in $C \setminus D$, obtaining C' . The rooted graph $H = R_{C',D}$ remains 4-light and of nonpositive density. This is the root-forgetting observation [2, Observation 5.2]: if a root set of order at most four is reduced, each forgotten root contributes at most four incident edges outside a given old fragment, offset by its density cost four. The roots of H lie in A , so $(A \cap D, B)$ is a root separation of H with right side $R_{A,B}$. Theorem 4.1 gives the helper model there. It contains one of the rooted graphs allowed in Lemma 2.5; that lemma supplies a reducible fragment of H . This is also reducible in G , since at least $|C'| \geq 3$ vertices remain outside it. Lemma 5.1 excludes it.

For (1), suppose that the side opposite a heavy side $R_{A,B}$ is not 4-light. It contains a positive fragment $L \subseteq A \setminus B$ with at most four neighbours. Bilightness makes $R_{A,B}$ 4-light, since a positive small-boundary fragment inside $B \setminus A$ would form a dense bfragment with L . Set

$$(C, D) = (L \cup N_G(L), V(G) \setminus L).$$

The side $R_{C,D}$ is 4-light and has nonpositive density: otherwise its whole interior, or a positive small-boundary fragment inside it, would again pair with L . Also $|L \cup N_G(L)| \geq 6$, since L has positive density. Now $B \subseteq D$, contradicting the preceding small-separation step.

Let $a \geq b$ be the two densities at a five-separation with boundary X . Then

$$(5) \quad a + b = \rho(G) + 20 - e(G[X]) \geq m + 8.$$

In particular, $a \geq \lceil (m+8)/2 \rceil$. If $b \geq 2$, (1) makes both sides 4-light. The rooted clique bound (Corollary 2.3) applies to the side of density a ; Theorem 4.1 applies to the other. Join matching root bags at their common root. The five joined bags form a clique; the two helper bags stay in the second open side and retain at least ten of their eleven possible contacts. This is a K_7^- model. Thus $b \leq 1$, and (5) gives $a \geq m+7$, proving (2).

For (3), first consider two heavy five-fragments S, T forming a bifragment. By (1), both R_S and R_T are 4-light. We claim that their closed sides can be joined by five disjoint paths. Otherwise Menger's theorem gives a separation (C, D) of order at most four containing $S \cup N_G(S)$ in C and $T \cup N_G(T)$ in D . Here $|C \setminus D| \geq 3$, since $|S| \geq 2$.

We verify that $R_{C,D}$ is 4-light and has nonpositive density. Discard from its root set the vertices with no neighbour in $D \setminus C$, leaving a separation (C, D') . No vertex of S can lie in $C \cap D'$, since all neighbours of S lie in C . Consequently $R_{C,D'}$ is a root side of order at most four in the side opposite R_S . By (1) it has nonpositive density, equal to that of $R_{C,D}$. Applying the same argument to every root side of $R_{C,D}$ proves its 4-lightness. The small-separation step, now applied to R_T , gives a contradiction. The five paths therefore exist.

Trim these paths so that they run between $N_G(S)$ and $N_G(T)$, with interiors outside both closed sides. The ends are distinct on each five-element boundary; a shared boundary vertex uses a trivial path. Apply the helper theorem in R_S and the rooted clique bound in R_T , using (2). Join corresponding root bags along these paths. Their interiors are disjoint and assigned to their own joined bag, so all bags remain disjoint and connected. Again they give K_7^- . This excludes two heavy five-fragments. A heavy five-fragment and a positive fragment of boundary at most four contradict (1). Two positive fragments of boundary at most four contradict bilightness. $\square$

Lemma 5.3. *We have $e(G) = 4v(G) - 2$.*

Proof. If $\rho(G) \geq -1$, delete an edge uv to obtain J . Minimality forces a dense bifragment (S, T) in J , with both boundaries of order at most four. Unless this already violates bilightness in G , we may label the ends so that $u \in S$ and $v \notin S \cup N_J(S)$. If $v \in T$, choose the labels also so that $|N_G(S)| \geq |N_G(T)|$. Then $|N_G(S)| \leq 5$ and $\rho_G(S) = \rho_J(S) + 1 \geq 2$, so $|S| \geq 2$.

Put $S' = S \setminus \{u\}$. We have

$$N_G(S') \subseteq (N_G(S) \setminus \{v\}) \cup \{u\}, \quad \rho_G(S') = \rho_G(S) + 4 - e_G(\{u\}, N_G(S)).$$

The pair (S', T) is nonadjacent in G . If $|N_G(S)| \leq 4$, both members have boundary at most four and positive density, a contradiction. Otherwise Lemma 5.2(2) gives $\rho_G(S) \geq 7$, hence $\rho_G(S') \geq 6$. The density of T is positive, and at least two if its boundary grows to order five. Lemma 5.2(3) again gives a contradiction. $\square$

6. DEGREE BOUNDS AND FIVE-VERTEX CUTS

Lemma 6.1. *The graph G has minimum degree at least five and is five-connected.*

Proof. If $d_G(v) \leq 4$, the smaller minor $J = G - v$ has density at least -2 and at least eight vertices. Any dense bifragment in J with boundaries of order at most four remains a bifragment in G . Its members remain positive; if a boundary grows to order five, the restored vertex adds at least one incident edge, so the density is at least two. Lemma 5.2(3) excludes such a pair. Thus J is 4-bilight, contrary to minimality.

Suppose that G has connectivity $k \leq 4$, and let S be a minimum cut, possibly empty. Each component of $G - S$ has neighbourhood S . By bilightness some component A has nonpositive density. The side R_A is internally k -connected, and its nonroots retain degree at least five. For $k \leq 2$, its connected interior gives a rooted clique directly. For $k = 3$, use Lemma 3.2; for $k = 4$, use Lemma 3.3. In the last case $|A| \geq 2$, since the minimum degree is five. Another component together with S has at least six vertices, again by the degree bound. Hence A is reducible, contrary to Lemma 5.1. $\square$

Write $t(xy) = |N_G(x) \cap N_G(y)|$ for an edge xy . Contracting it changes the global density by

$$(6) \quad \rho(G/xy) = \rho(G) + 3 - t(xy).$$

A quasi-five-connected graph is 4-bilight: any member of a dense bifragment has boundary four and at least two vertices, and the other member supplies at least two vertices on the opposite side of that cut. Thus a quasi-five-connected contraction with $t(xy) \leq 3$ would contradict minimality.

We will use one further observation repeatedly. In a five-connected graph, contraction of an edge gives a four-connected graph, and every four-cut contains the merged vertex. Consequently, in a dense bifragment of the contraction with boundaries of order at most four, the merged vertex belongs to both boundaries. Its open sets lift to positive five-fragments of the original graph, with both contracted ends in each boundary. Their densities do not decrease; they remain disjoint and nonadjacent. Each has at least two vertices, because a singleton of boundary four has density zero in the contraction.

Lemma 6.2. *The minimum degree of G is at least six.*

Proof. Suppose that $d_G(v) = 5$. If $N_G(v)$ is a clique, choose a component C outside $N_G[v]$. Such a component exists because $v(G) \geq 9$, and five-connectivity gives $N_G(C) = N_G(v)$. The five singleton neighbours, $\{v\}$ and C form a K_7^- model.

Otherwise choose nonadjacent $a, b \in N_G(v)$ and set $J = G - v + ab$. This is a minor: contract va , then delete the new edges at a other than ab . It has density -2 , so a dense bifragment (Y, Z) with boundaries of order at most four exists in J . For either member, say Y , put $k = e_G(\{v\}, Y)$ and let c be one if the added edge ab has an end in Y , and zero otherwise. Then

$$(7) \quad \rho_G(Y) = \rho_J(Y) + k - c.$$

Since $c = 1$ implies $k \geq 1$, both sets remain positive in G ; they are still disjoint and nonadjacent, and their boundaries have order at most five. Lemma 5.2(3) therefore lets us choose Y with boundary five and $\rho_G(Y) = 1$. Its boundary grows on restoring v , so $k \geq 1$. Equation (7) forces $\rho_J(Y) = k = c = 1$. Exchange a, b if necessary: a is the unique neighbour of v in Y , and b lies in the boundary of Y in both graphs. Hence $|Y| \geq 2$. The opposite side of Y in G contains Z , which has at least two vertices.

If $|Y| \geq 3$, Lemma 3.4 makes G/va quasi-five-connected. As ab is absent and $d_G(v) = 5$, we have $t(va) \leq 3$, a contradiction. We are left with $Y = \{a, c\}$. The equation

$$1 = d_G(a) + d_G(c) - e(G[\{a, c\}]) - 8$$

forces $ac \in E(G)$ and $d_G(a) = d_G(c) = 5$. Their common neighbours form a triple R , and their complete neighbourhoods are

$$(8) \quad N_G(a) = \{c, v\} \cup R, \quad N_G(c) = \{a, b\} \cup R, \quad |R| = 3.$$

In particular vb is an edge.

Contract ac , which has exactly three common neighbours. By minimality and the preceding contraction observation, a dense bifragment lifts to two positive five-fragments. Choose a member T of density one, using Lemma 5.2(3). Lifting increases its density by $|T \cap R|$, so T avoids R . Both a, c are in its boundary; thus $v, b \in T$, and v is the unique neighbour of a in T . Its opposite side has at least two vertices. If $|T| \geq 3$, Lemma 3.4 applies to av ; the missing edge cv gives $t(av) \leq 3$, again a contradiction. Therefore $T = \{v, b\}$, and the same degree calculation gives

$$(9) \quad N_G(v) = \{a, b\} \cup D, \quad N_G(b) = \{c, v\} \cup D, \quad |D| = 3.$$

The four degree-five vertices induce the cycle a, c, b, v, a , and R, D avoid the cycle.

If $|R \cap D| \geq 2$, the cycle has at most four external neighbours. There is a vertex outside the cycle and its boundary, because $v(G) \geq 9$, contradicting five-connectivity. If $|R \cap D| = 1$, put $S = R \cup D$ and $U = \{a, c, b, v\}$. Here $|S| = 5$ and $\rho_G(U) = 0$. Its opposite side W has

$$\rho_G(W) = 18 - e(G[S]) \geq 8.$$

In particular W is nonempty. Its closed side rooted at S is internally five-connected, so the rooted clique theorem applies. The connected bags $\{a, v\}$ and $\{c, b\}$ are adjacent to each other and to every vertex of S . Together with the rooted clique they give a K_7 model.

Finally suppose that $R \cap D = \emptyset$. Every five-cut L containing a, v must also contain c, b . Indeed, if $c \notin L$, all of R lies in its component of $G - L$ or in L . Then a has no neighbour in any other component, and removing a from L leaves a cut of order four. The argument for b is the same, using the neighbours of v .

Now $t(av) = 0$. The contraction G/av has sufficient density, so it has a dense biframe. Both lifted five-boundaries contain all four cycle vertices, by the preceding paragraph. Each open side meets both R and D , because its boundary contains a, v . One of these disjoint sides, say B , contains exactly one vertex r of R . It cannot be a singleton, since its image has positive density and boundary four. If $|B| = 2$, then $B = \{r, d\}$ for some $d \in D$, and its boundary consists of the four cycle vertices and one further vertex z . The vertex r is adjacent to neither v nor b , so its neighbours would all lie in $\{a, c, d, z\}$, contrary to the minimum degree five.

Thus $|B| \geq 3$, its opposite side has at least two vertices, and a has the unique neighbour r in B . Lemma 3.4 makes G/ar quasi-five-connected. Also $t(ar) \leq 3$, since a has degree five and its neighbour v is not adjacent to r . This final contraction contradicts minimality. All contractions used above leave at least eight vertices, so the order condition is retained throughout. $\square$

Lemma 6.3. *Every proper low five-fragment A contains an endpoint of an edge e with $t(e) \leq 3$.*

Proof. Put $S = N_G(A)$ and $a = |A|$. The minimum degree six gives

$$\rho_G(A) = \sum_{v \in A} d_G(v) - e(G[A]) - 4a \geq 2a - \binom{a}{2},$$

so $a \geq 5$. Also

$$\sum_{v \in A} (d_G(v) + |N_G(v) \cap S|) = 8a + 2\rho_G(A) \leq 8a + 2.$$

Choose $v \in A$ whose summand is at most eight. Suppose that every edge at v has at least four common neighbours. Let H be $G[N_G(v)]$ with the edges within $X_0 = N_G(v) \cap S$ deleted. Vertices of $H - X_0$ have degree at least four, and vertices in X_0 have degree at least $5 - |X_0|$. Moreover $v(H) + |X_0| \leq 8$.

The side $G[A \cup S]$, rooted at S , is internally five-connected. Choose five disjoint paths from S to $N_G(v)$, stopped when they first reach $N_G(v)$. They avoid v , and the paths at vertices of X_0 are trivial. Their five ends form a set containing X_0 . Apply Lemma 3.1 to H rooted at these ends. Its central bag avoids S . Use this bag and $\{v\}$ as helpers. Four of the linkage paths become root bags; delete the last vertex of the central path to obtain the fifth root bag. This trimmed path is nonempty, since its end lies outside X_0 and hence outside S . Both helpers meet the other four root bags, the helpers are adjacent, and the trimmed bag meets the central helper. Thus at least ten of the eleven helper contacts are present. Every bag lies in $A \cup S$, and no artificial boundary edge has been used.

The opposite side is internally five-connected and has density at least $m + 7$, where $m = 10 - e(G[S])$, by Lemma 5.2(2). It supplies a rooted clique on S . Joining corresponding root bags gives K_7^- , a contradiction. $\square$

The following elementary end argument will exclude the remaining five-cuts. A proper five-fragment is called an F -fragment, for a family F of edges, if its boundary contains both ends of an edge in F . An F -end is an F -fragment containing no F -fragment as a strict subset.

Lemma 6.4. *Let J be five-connected, with every proper five-fragment of order at least four. If A is an F -end, no edge $e \in F$ can meet A , lie in $A \cup N_J(A)$ and have both ends in a five-cut of J .*

Proof. Put $S = N_J(A)$ and $C = V(J) \setminus (A \cup S)$. Suppose a five-cut T contains both ends of such an edge e , and partition $J - T$ into two nonempty unions of components B, D . Write

$$\begin{aligned} p &= |A \cap T|, & s &= |S \cap T|, & r &= |C \cap T|, \\ u &= |S \cap B|, & w &= |S \cap D|. \end{aligned}$$

Then $p + s + r = u + s + w = 5$, $p \geq 1$ and $p + s \geq 2$.

If $A \cap B \neq \emptyset$, its neighbourhood lies in $(A \cap T) \cup (S \cap T) \cup (S \cap B)$. This containing set cannot have order below five, by connectivity. If it has order five, it equals the actual boundary and contains both ends of e , giving an F -fragment strictly inside A . Its opposite side contains C , and strict containment follows from $p \geq 1$. Therefore

$$p + s + u \geq 6, \quad p > w, \quad u > r.$$

The opposite corner $C \cap D$ has boundary contained in a set of order $r + s + w = 5 + r - u < 5$, so it is empty. Symmetrically, if $A \cap D \neq \emptyset$, then $p > u$, $w > r$ and $C \cap B$ is empty. In these applications a nonempty opposite corner witnesses that the candidate separation is proper.

If both A -corners are nonempty, then $C \subseteq T$, so $r = |C| \geq 4$, contrary to $p + s \geq 2$. If just $A \cap B$ is nonempty, then $D = S \cap D$, giving $w = |D| \geq 4$. Since $p > w$, we have $p = 5$ and $r = s = 0$. Now $C = C \cap B$ has all its neighbours in $S \cap B$, of order $u = 5 - w \leq 1$, again impossible. The other one-corner case is symmetric.

If neither A -corner is nonempty, then $A \subseteq T$ and $p = |A| \geq 4$. Since $|C| \geq 4$ and $r \leq 1$, a C -corner is nonempty, say $C \cap B$. Its boundary has order at most $r + s + u$, so connectivity gives $u \geq p \geq 4$. The other C -corner must be empty, or similarly $w \geq 4$. But then $D = S \cap D$, and again $w = |D| \geq 4$. This contradicts $u + w \leq 5$ in every case. $\square$

Lemma 6.5. *The graph G is six-connected.*

Proof. It is already five-connected. We first show that every proper five-fragment has at least four vertices. A singleton would have degree five. For a fragment of order $a = 2$ or 3 , the degree bound and the possible incident edges give

$$2a - \binom{a}{2} \leq \rho_G(A) \leq \binom{a}{2} + a.$$

Its density is therefore three if $a = 2$, and between three and six if $a = 3$. Both contradict Lemma 5.2(2), which allows only densities at most one or at least seven. Notice also that a low five-fragment is automatically proper: an empty opposite side would give density $18 - e(G[N_G(A)]) \geq 8$.

Let F be the edges with at most three common neighbours. Every edge in F has both ends in the boundary of a five-fragment of density one. Indeed, its contraction preserves the density bound by (6), so is not 4-bilight. Its dense bifragment lifts to two positive five-fragments, as described above; Lemma 5.2(3) forces one density to equal one.

Suppose a five-cut exists. It has a low side, so Lemma 6.3 supplies an edge of F . The preceding paragraph supplies a low F -fragment. Choose an inclusion-minimal one, A . In fact A is an F -end without any density restriction. For a smaller F -fragment $B \subsetneq A$, a heavy B would be disjoint and nonadjacent to the heavy opposite side of A , violating Lemma 5.2(3). Thus B would be low, contradicting the choice of A .

Lemma 6.3 now supplies an edge $e \in F$ meeting A . Its other end lies in $A \cup N_G(A)$, and both ends lie in a five-cut by the earlier contraction argument. This contradicts Lemma 6.4 and excludes every five-cut. $\square$

7. THE FINAL DEGREE CASES

The degree-six case uses the two-paths theorem of Robertson, Seymour and Thomas [6, Theorem (2.4)], in the form also quoted in [5, Theorem 13]. For four distinct vertices in a specified cyclic order, it gives disjoint paths joining opposite pairs, a separation of order at most three with all terminals on one side and at least two vertices in the other open side, or a drawing in a disc with the terminals on its boundary in that order.

Lemma 7.1. *Let H be six-connected, with at least nine vertices and no K_7^- minor. If every edge has at least four common neighbours and H has a vertex of degree six, then $e(H) \leq 4v(H) - 9$.*

Proof. Let $d_H(v) = 6$. The complement of $H[N_H(v)]$ is a matching, since each neighbour of v has at least four neighbours there. Label the six neighbours $u_1, w_1, u_2, w_2, u_3, w_3$ so that the only possible nonedges are $u_i w_i$.

There cannot be disjoint paths joining two different pairs u_i, w_i with interiors outside $N_H[v]$: contracting the interior of each path into one of its ends fills two of the three possible missing contacts. Together with $\{v\}$, the six resulting bags give K_7^- . In particular all three pairs are nonedges. Otherwise an existing pair edge and a path for another pair through an exterior component would suffice. Such a component exists and is adjacent to all six neighbours of v , by six-connectivity.

Some pair, say u_1, w_1 , has no common neighbour outside $N_H[v]$. Otherwise choose an exterior common neighbour x_i for each pair. Two distinct choices give the forbidden disjoint length-two paths, so all choices coincide at a vertex x . There is a component C outside $N_H[v] \cup \{x\}$, because $v(H) \geq 9$. Its boundary is contained in $N_H(v) \cup \{x\}$ and has at least six vertices. Thus at least five neighbours of v meet C , giving a path for one pair through C and a disjoint path for another through x .

Apply the two-paths theorem to $H' = H - \{v, u_1, w_1\}$, with ordered terminals u_2, u_3, w_2, w_3 . The two paths are excluded above. A separation (A, B) in the second outcome lifts to

$$(A \cup \{v, u_1, w_1\}, B \cup \{u_1, w_1\})$$

in H , of order at most five. It is proper: v lies in the first open side, and the second contains $B \setminus A$. No neighbour of v lies in that second open side. This contradicts six-connectivity.

Hence H' has the disc drawing. Its four distinct boundary terminals give $e(H') \leq 3v(H') - 7$. Every vertex outside $N_H[v]$ meets the deleted triple at most once, by the choice of u_1, w_1 . Each of the other four neighbours of v meets it three times, and the triple induces two edges. With $n' = v(H') = v(H) - 3$,

$$e(H) \leq (3n' - 7) + (n' - 4) + 12 + 2 = 4v(H) - 9.$$

□

Proof of Theorem 2.1. Our minimum counterexample G is six-connected by Lemma 6.5. Every edge has at least four common neighbours: otherwise contracting it gives a five-connected, hence 4-bilight, smaller graph of density at least -2 . The average degree is $8 - 4/v(G) < 8$, whereas Lemma 6.2 gives minimum degree at least six. Thus some vertex has degree six or seven.

Degree six contradicts Lemma 7.1, since $e(G) = 4v(G) - 2$. For a degree-seven vertex v , put $J = G[N_G(v)]$. The common-neighbour bound gives $\delta(J) \geq 4$. Choose a component C outside $N_G[v]$, which exists since $v(G) \geq 9$. Its boundary lies in $N_G(v)$ and has at least six vertices. Contract C to a vertex and delete the other exterior components. The resulting nine-vertex minor consists of J , the vertex v adjacent to all of J , and a nonadjacent second vertex adjacent to at least six vertices of J . Lemma A.1 gives a K_7^- model in this minor. Replacing the contracted vertex by the connected set C lifts the model to G .

Both degree cases are impossible, so no counterexample exists. □

Proofs of Corollary 1.2 and Theorem 1.1. Every five-connected graph is 4-bilight, giving Corollary 1.2. For the colouring theorem, suppose a K_7^- -minor-free graph is not six-colourable, and choose such a graph G minor-minimal. Then its chromatic number is at least seven and every proper minor is six-colourable. Dvořák, Norin and Rahman [2, Theorem 1.6] give seven-connectivity and $e(G) \geq 4v(G) - 2$. The graph is therefore 4-bilight, and Theorem 2.1 supplies the forbidden minor. □

APPENDIX A. THE FINITE DEGREE-SEVEN LEMMA

Lemma A.1. *Let R be a graph on seven vertices with minimum degree at least four. Add a vertex v adjacent to every vertex of R , and another vertex c adjacent to at least six vertices of R . No edge vc is required. The resulting graph contains a K_7^- minor.*

This lemma is computer-assisted. Its verification concerns only nine-vertex graphs; the global proof supplies the reduction from hosts of arbitrary order. We describe the enumeration and the certificates so that the scope of the computation is explicit.

The complement F of R has maximum degree at most two. Each component of F is therefore a path, an isolated vertex, or a cycle of length at least three. Enumerating multisets of these

components with total order seven gives 29 isomorphism types. For each type, the verifier tests all eight possibilities for c : it meets every vertex of R , or misses one specified vertex. Thus there are $29 \cdot 8 = 232$ cases. Extra edges, including vc , can be discarded.

For each case, the program searches for seven bags among the nine vertices. It enumerates every partition of a subset of order seven, eight or nine into seven nonempty blocks. There are respectively 36, 252 and 462 such partitions. A certificate records the bags as vertex bitmasks. The verifier checks that they are disjoint and nonempty, that every bag is connected, and that at most one of the 21 pairs of bags lacks an edge. These checks certify a K_7^- minor directly; no minor-testing library is used.

All 232 cases pass. The returned models use seven vertices in 67 cases, eight in 102 cases and nine in 63 cases. The program also checks a literal K_7^- as a positive example and a padded K_6 and $K_{2,2,2,2}$ as negative examples. It validates each returned certificate after the search; this validation reuses the connectivity and contact predicates and is not an independently implemented checker.

The complete standard-library Python program is preserved at a fixed revision in the accompanying repository:

`hc7_k7minus_degree7_quotient_verify.py`.

Run it with `uv run python3` from a checkout of that revision. The manuscript’s accompanying README records the source hash, expected output and certificate digest.

USE OF AI

OpenAI Codex was used for mathematical exploration, computation, drafting and internal proof checks. This draft has not undergone external mathematical review.

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